Math for CS & AI: Homework 10

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Problem 5.9

In Definition 2.96, we defined the independence of random variables. Although the definition is stated only for the finite discrete case, it is also the general definition of independence for arbitrary random variables.

(a) Prove that if $(X, Y) \in \mathbb{R}^2$ is a Gaussian random vector, then X and Y are independent if and only if $\mathrm{Cov}(X, Y) = 0$.

(b) Let $(X, Y) \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ be a standard Gaussian random vector in $\mathbb{R}^2$. Are the following two random variables independent?

$$X + Y \quad \text{and} \quad X - Y.$$

Answer:

For part (a), if X and Y are independent, then $\mathrm{Cov}(X, Y) = 0$ by a basic property of covariance. If $\mathrm{Cov}(X, Y) = 0$, then the covariance matrix is diagonal, which implies there exists a diagonal matrix $\boldsymbol{A} \in \mathbb{R}^{2\times 2}$, a vector $\boldsymbol{\mu} \in \mathbb{R}^2$, and a standard Gaussian vector $\boldsymbol{z} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_2)$ such that $(X, Y) = \boldsymbol{\mu} + \boldsymbol{A}\boldsymbol{z}$. Therefore, X is a function only of $\boldsymbol{\mu}_1$, $\boldsymbol{A}_{11}$, and $\boldsymbol{z}_1$, while Y is a function only of $\boldsymbol{\mu}_2$, $\boldsymbol{A}_{22}$, and $\boldsymbol{z}_2$. Since $\boldsymbol{z}_1$ and $\boldsymbol{z}_2$ are independent standard Gaussian random variables, X and Y are independent too.

For part (b), the random variables $X + Y$ and $X - Y$ are indeed independent. This follows directly from computing their covariance:

$$\begin{aligned}
\mathrm{Cov}(X + Y, X - Y) &= \mathbb{E}[(X + Y - \mathbb{E}[X + Y])(X - Y - \mathbb{E}[X - Y])] \\
&= \mathbb{E}[(X - \mathbb{E}[X] + Y - \mathbb{E}[Y])(X - \mathbb{E}[X] - Y + \mathbb{E}[Y])] \\
&= \mathbb{E}[(X - \mathbb{E}[X])^2 - (Y - \mathbb{E}[Y])^2] \\
&= \mathrm{Var}(X) - \mathrm{Var}(Y) = 1 - 1 = 0,
\end{aligned}$$

since X and Y are independent standard Gaussian random variables with $\mathrm{Var}(X) = \mathrm{Var}(Y) = 1$. By the result from part (a), this means $X + Y$ and $X - Y$ are independent.

Problem 5.10

In general, let X, Y be two random variables on the same probability space. Let $g(A, y)$ be a function that maps a set $A \subseteq \mathbb{R}$ and a real number $y \in \mathbb{R}$ to a real number. We say that $g(A, y)$ is a version of the conditional probability distribution $P_{X|Y}$ if it satisfies that for every $A, B \subseteq \mathbb{R}$ with $\Pr[Y \in B] > 0$,

$$\Pr[X \in A \mid Y \in B] = \mathbb{E}[g(A, Y) \mid Y \in B].$$

Please prove the following statements:

(a) Let $(X, Y) \in \mathbb{R}^2$ be a random vector with PDF $f(x, y)$. Show that the following function is a version of $P_{X|Y}$:

$$g(A, y) := \frac{\int_A f(x, y)\,\mathrm{d}x}{\int_{\mathbb{R}} f(x, y)\,\mathrm{d}x}.$$

Here, when the denominator is zero, we define $g(A, y)$ arbitrarily.

(b) Let $(X, Z) \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ be a standard Gaussian random vector in $\mathbb{R}^2$. Let $Y := X + Z$. Determine a version of $P_{X|Y}$.

(c) Similar to probability density functions, the conditional probability distributions may not be unique. In the setting of the previous part, construct another version of $P_{X|Y}$.

Note: In fact, the conditional probability distribution is unique in the following sense: if X and Y are two random variables on the same probability space, and g, g' are two versions of $P_{X|Y}$, then there exists a set $B \subseteq \mathbb{R}$ with $\Pr[Y \in B] = 1$ such that for all $A \subseteq \mathbb{R}$ and $y \in B$,

$$g(A, Y) = g'(A, Y).$$

Answer:

For part (a), we verify this directly by definition. Let $f_Y(y) = \int_{\mathbb{R}} f(x, y)\,\mathrm{d}x$ denote the PDF of Y, then

$$\begin{aligned}
\Pr[X \in A \mid Y \in B] &= \frac{\Pr[X \in A, Y \in B]}{\Pr[Y \in B]} \\
&= \frac{\int_B \int_A f(x, y)\,\mathrm{d}x\,\mathrm{d}y}{\int_B f_Y(y)\,\mathrm{d}y}, \\
\mathbb{E}[g(A, Y) \mid Y \in B] &= \frac{\mathbb{E}\left[g(A, Y) \cdot \mathbb{1}_{[Y \in B]}\right]}{\Pr[Y \in B]} \\
&= \frac{\int_B g(A, y) f_Y(y)\,\mathrm{d}y}{\int_B f_Y(y)\,\mathrm{d}y}.
\end{aligned}$$

Substituting $g(A, y) = \frac{\int_A f(x,y)\,\mathrm{d}x}{f_Y(y)}$, we derive that

$$\Pr[X \in A \mid Y \in B] = \mathbb{E}[g(A, Y) \mid Y \in B] = \frac{\int_B \int_A f(x, y)\,\mathrm{d}x\,\mathrm{d}y}{\int_B f_Y(y)\,\mathrm{d}y}.$$

For part (b), we have $X = \frac{1}{2}Y + \frac{1}{2}(X - Z)$. From Problem 5.9(b), we know that $Y = X + Z$ is independent of $X - Z$. Since $X - Z \sim \mathcal{N}(0, 2)$, given $Y = y$, the conditional distribution of X is $\mathcal{N}\left(\frac{y}{2}, \frac{1}{2}\right)$. Therefore, we

Problem 5.10 continued on next page...

can construct a version of $P_{X|Y}$:

$$g(A, y) = \int_A \frac{1}{\sqrt{2\pi \cdot 1/2}} e^{-(x-y/2)^2/(2\cdot 1/2)} \, \mathrm{d}x = \int_A \frac{1}{\sqrt{\pi}} e^{-(x-y/2)^2} \, \mathrm{d}x.$$

For part (c), we construct another version of $P_{X|Y}$:

$$g'(A, y) = \begin{cases} 1, & \text{if } y = 0 \text{ and } 0 \in A, \\ 0, & \text{if } y = 0 \text{ and } 0 \notin A, \\ g(A, y), & \text{otherwise.} \end{cases}$$

Since Y is a continuous random variable, $\Pr[Y = 0] = 0$. Modifying the value at the single point $y = 0$ does not affect the integral over any set $B \subseteq \mathbb{R}$ with $\Pr[Y \in B] > 0$.

Problem 5.11

Let $\boldsymbol{\mu} \in \mathbb{R}^d$ and $\boldsymbol{\Sigma} \in \mathbb{R}^{d \times d}$. Show that $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is a valid Gaussian distribution on $\mathbb{R}^d$ if and only if $\boldsymbol{\Sigma}$ is symmetric and positive semi-definite. Here a matrix $\boldsymbol{\Sigma}$ is called positive semi-definite if for all $\boldsymbol{x} \in \mathbb{R}^d$, we have $\boldsymbol{x}^\top \boldsymbol{\Sigma} \boldsymbol{x} \geq 0$. According to our definition in Definition 5.35 (other textbooks may use different definitions), a Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is valid if there exists $\boldsymbol{A} \in \mathbb{R}^{d \times r}$ for some r such that $\boldsymbol{\Sigma} = \boldsymbol{A}\boldsymbol{A}^\top$.

Hint: Any real symmetric matrix $\boldsymbol{A}$ can be diagonalized as $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\top$, where $\boldsymbol{\Lambda}$ is a diagonal matrix and $\boldsymbol{Q}$ is an orthogonal matrix. The diagonal entries of $\boldsymbol{\Lambda}$ are the eigenvalues of $\boldsymbol{A}$.

Answer:

If there exists a matrix $\boldsymbol{A} \in \mathbb{R}^{d \times r}$ such that $\boldsymbol{\Sigma} = \boldsymbol{A}\boldsymbol{A}^\top$, then $\boldsymbol{\Sigma}^\top = (\boldsymbol{A}\boldsymbol{A}^\top)^\top = (\boldsymbol{A}^\top)^\top \boldsymbol{A}^\top = \boldsymbol{A}\boldsymbol{A}^\top = \boldsymbol{\Sigma}$, which shows that $\boldsymbol{\Sigma}$ is symmetric. For any vector $\boldsymbol{x} \in \mathbb{R}^d$, we have $\boldsymbol{x}^\top \boldsymbol{\Sigma} \boldsymbol{x} = \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{A}^\top \boldsymbol{x} = (\boldsymbol{A}^\top \boldsymbol{x})^\top (\boldsymbol{A}^\top \boldsymbol{x}) = \left\|\boldsymbol{A}^\top \boldsymbol{x}\right\|_2^2 \geq 0$, which shows that $\boldsymbol{\Sigma}$ is positive semi-definite.

If $\boldsymbol{\Sigma}$ is symmetric and positive semi-definite, then we can diagonalize it as $\boldsymbol{\Sigma} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\top$, where $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_d)$ and $\boldsymbol{Q}$ is an orthogonal matrix. Since $\boldsymbol{\Sigma}$ is positive semi-definite, for each unit eigenvector $\boldsymbol{q}_i$ corresponding to eigenvalue λ_i, we have $\boldsymbol{q}_i^\top \boldsymbol{\Sigma} \boldsymbol{q}_i = \lambda_i \geq 0$. Let $\boldsymbol{\Lambda}^{1/2} = \mathrm{diag}(\sqrt{\lambda_1}, \sqrt{\lambda_2}, \ldots, \sqrt{\lambda_d})$. Since $\boldsymbol{\Lambda}^{1/2}$ is diagonal, its transpose is equal to itself, so $\boldsymbol{\Lambda} = \boldsymbol{\Lambda}^{1/2}(\boldsymbol{\Lambda}^{1/2})^\top$. Therefore, we can construct the matrix $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}^{1/2}$, and then $\boldsymbol{A}\boldsymbol{A}^\top = \boldsymbol{Q}\boldsymbol{\Lambda}^{1/2}(\boldsymbol{\Lambda}^{1/2})^\top \boldsymbol{Q}^\top = \boldsymbol{\Sigma}$.

Problem 5.12

Let $B^d := \{x \in \mathbb{R}^d : \|x\|_2 \leq 1\}$ be the unit ball in $\mathbb{R}^d$. Consider an axis-aligned hypercube of the following form:

$$H := [\ell_1, u_1] \times [\ell_2, u_2] \times \cdots \times [\ell_d, u_d], \tag{5.312}$$

where $u_1 - \ell_1 = u_2 - \ell_2 = \cdots = u_d - \ell_d > 0$.

(a) Among all such hypercubes H that are contained in B^d, what is the maximum possible volume of H?

(b) Let V_d be the volume of B^d and U_d be the maximum possible volume of H that is contained in B^d. What is the limit of $\frac{U_d}{V_d}$ as $d \to \infty$?

Hint: You can use the fact that the volume of B^d is

$$V_d = \frac{\pi^{d/2}}{\Gamma(d/2+1)} = \Theta\left(\frac{1}{\sqrt{d}}\left(\frac{2\pi e}{d}\right)^{d/2}\right),$$

where $\Gamma(x) := \int_0^\infty t^{x-1}e^{-t}\,dt$ is the Gamma function

Answer:

For part (a), we first argue that the optimal hypercube must be centered at the origin, i.e., $|\ell_i| = |u_i|$ for all i. This is because the unit ball is symmetric about the origin, and for any non-centered hypercube, we can always shift and expand it to obtain a centered hypercube with larger volume that still lies within the unit ball. For a centered hypercube, the vertices farthest from the origin have coordinates $(\pm u_1, \pm u_2, \ldots, \pm u_d)$, so the containment condition becomes $\sum_{i=1}^d u_i^2 \leq 1$. The volume of H is $\prod_{i=1}^d (u_i - \ell_i) = 2^d \prod_{i=1}^d u_i$. By the AM-GM inequality, the product $\prod_{i=1}^d u_i$ is maximized when all u_i are equal, subject to the constraint $\sum_{i=1}^d u_i^2 = 1$. Therefore, we choose $u_i = \frac{1}{\sqrt{d}}$ for all i, and the maximum volume is $U_d = \left(\frac{2}{\sqrt{d}}\right)^d$.

For part (b), using the asymptotic formula for V_d given in the hint, we have

$$\frac{U_d}{V_d} = \frac{\left(\frac{2}{\sqrt{d}}\right)^d}{\Theta\left(\frac{1}{\sqrt{d}}\left(\frac{2\pi e}{d}\right)^{d/2}\right)} = \Theta\left(\sqrt{d}\cdot\left(\frac{2}{\pi e}\right)^{d/2}\right) \to 0 \quad \text{as } d \to \infty.$$

Since $\frac{2}{\pi e} < 1$, the exponential term decays to zero much faster than $\sqrt{d}$ grows. Therefore, the limit is 0.

Problem 5.13

Let $\boldsymbol{x} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}/d)$. In Theorem 5.43, we showed that $|\|\boldsymbol{x}\|_2^2 - 1| \geq \epsilon$ with probability at most $2e^{-d\epsilon^2/8}$, but this only holds for $\epsilon \in (0, 1)$.

(a) Prove that for all $\epsilon \geq 1$,
$$\Pr\left[\left|\|\boldsymbol{x}\|_2^2 - 1\right| \geq \epsilon\right] \leq e^{-d\epsilon/8}.$$

(b) Is it possible to improve the above upper bound to $\leq e^{-Cd\epsilon^2}$ for some constant $C > 0$?

(c) Prove that for all $\epsilon > 0$,
$$\Pr\left[|\|\boldsymbol{x}\|_2 - 1| \geq \epsilon\right] \leq 2e^{-d\epsilon^2/8}.$$

Answer:

For part (a), the same Chernoff bound argument used to prove Theorem 5.43 can be applied here. We derived that
$$\forall a \geq 1, \quad \Pr[\|\boldsymbol{x}\|_2^2 \geq a] \leq e^{-dI_{Z^2}(a)},$$
where $I_{Z^2}(a) = \frac{1}{2}(a - 1 - \log a)$ is the rate function of Z^2, where $Z \sim \mathcal{N}(0, 1)$ is a standard Gaussian random variable. For $\epsilon \geq 1$, note that $1 - \epsilon \leq 0$ and $\|\boldsymbol{x}\|_2^2 \geq 0$ always holds, so the event $\left|\|\boldsymbol{x}\|_2^2 - 1\right| \geq \epsilon$ is equivalent to $\|\boldsymbol{x}\|_2^2 \geq 1 + \epsilon$. For every $\epsilon \geq 1$, we have
$$I_{Z^2}(1 + \epsilon) = \frac{1}{2}(\epsilon - \log(1 + \epsilon)).$$
We only need to prove that $x - \log(1 + x) \geq \frac{x}{4}$ for all $x \geq 1$. Let $f(x) = \frac{3}{4}x - \log(1 + x)$, taking the derivative gives $f'(x) = \frac{3}{4} - \frac{1}{1+x} > 0$ for all $x \geq 1$. Combining the fact that $f(1) = \frac{3}{4} - \ln 2 > 0$, the inequality holds. Therefore,
$$\Pr\left[\left|\|\boldsymbol{x}\|_2^2 - 1\right| \geq \epsilon\right] \leq e^{-d\epsilon/8}.$$

For part (b), it is impossible. The rate function $I_{Z^2}(1 + \epsilon) = \frac{1}{2}(\epsilon - \log(1 + \epsilon))$ satisfies $I_{Z^2}(1 + \epsilon) \sim \frac{\epsilon}{2}$ as $\epsilon \to \infty$. The Chernoff bound is tight in the exponential sense, so the tail probability decays as $e^{-\Theta(d\epsilon)}$, not $e^{-\Theta(d\epsilon^2)}$. Thus, no such constant $C > 0$ exists.

For part (c), for $\epsilon \in (0, 1)$, the result follows directly from Theorem 5.43. If $\epsilon \geq 1$, the event $|\|\boldsymbol{x}\|_2 - 1| \geq \epsilon$ is equivalent to $\|\boldsymbol{x}\|_2 \geq 1 + \epsilon$. Therefore, for $\epsilon \geq 1$, we have
$$\Pr\left[|\|\boldsymbol{x}\|_2 - 1| \geq \epsilon\right] = \Pr\left[\|\boldsymbol{x}\|_2^2 \geq (1 + \epsilon)^2\right] \leq e^{-d((1+\epsilon)^2-1)/8} = e^{-d(2\epsilon+\epsilon^2)/8} \leq 2e^{-d\epsilon^2/8}.$$

Problem 5.14

How can we sample from the standard Gaussian distribution $\mathcal{N}(0,1)$? According to Exercise 5.6, it seems that we only need to generate samples from U and then apply the quantile function of $\mathcal{N}(0,1)$ to them. But the quantile function of $\mathcal{N}(0,1)$ is not an elementary function, so it is not easy to compute it in a computer program. But remember: integrating one Gaussian is too hard, but integrating two Gaussians is easy.

Let $Z_1, Z_2 \sim U$, where U denotes the uniform distribution on $[0,1)$ as in Exercise 5.6. Construct a function $g : [0,1]^2 \to \mathbb{R}^2$ such that

$$g(Z_1, Z_2) \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}).$$

The function should only involve the following elementary operations and their compositions: addition, subtraction, multiplication, division, square root, exponential, logarithm, and trigonometric functions.

Answer:

To generate two independent standard Gaussian random variables $X, Y \sim \mathcal{N}(0,1)$, we use the polar coordinate transformation $x = r\cos\theta$ and $y = r\sin\theta$. We now generate the random variables R and Θ separately, which are independent. Generating Θ is straightforward: we can set $\Theta = 2\pi Z_1$.

The PDF of R is $f_R(r) = re^{-r^2/2}$ for $r \geq 0$. Then the CDF of R is

$$F_R(r) = \Pr[R \leq r] = \int_0^r te^{-t^2/2}\mathrm{d}t = 1 - e^{-r^2/2}, \quad \forall r \geq 0.$$

By the result from Problem 5.6, we can generate R as

$$R = F_R^{-1}(Z_2) = \inf\{r \geq 0 : 1 - e^{-r^2/2} \geq Z_2\} = \sqrt{-2\ln(1 - Z_2)}.$$

Therefore, we can construct the desired function $g : [0,1]^2 \to \mathbb{R}^2$ by

$$g(Z_1, Z_2) = \begin{bmatrix} \sqrt{-2\ln(1-Z_2)}\cos(2\pi Z_1) \\ \sqrt{-2\ln(1-Z_2)}\sin(2\pi Z_1) \end{bmatrix}.$$

Problem 5.15

Given n points $x_1, \ldots, x_n \in \mathbb{R}^d$, define the diameter of this point cloud as

$$\mathrm{diam}(x_1, \ldots, x_n) := \max_{i,j \in [n]} \|x_i - x_j\|_2.$$

A naïve algorithm to compute the diameter is to compute the distance between every pair of points, and then take the maximum. This takes $O(n^2 d)$ time because there are $O(n^2)$ pairs of points and each pair takes $O(d)$ time to compute the distance.

Describe an algorithm that inputs a point cloud $x_1, \ldots, x_n \in \mathbb{R}^d$ and parameters $\delta \in (0,1)$, $\epsilon \in (0,1)$, and satisfies:

- It always runs in time

$$\mathcal{O}\left(\frac{n(n+d)}{\epsilon^2}\log\frac{n}{\delta}\right).$$

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- With probability at least $1-\delta$, it outputs a pair of points (x_i, x_j) such that

$$\|x_i - x_j\|_2 \geq (1-\epsilon) \cdot \operatorname{diam}(x_1, \ldots, x_n).$$

Your algorithm should only involve elementary operations on integers and real numbers. Each elementary operation takes at most $O(1)$ time. Your algorithm can access a random number generator that generates samples from the uniform distribution U on $[0,1)$, and each sample takes at most $O(1)$ time to generate.

Note: If you wonder if this running time is optimal, the answer is no. It can be further reduced to

$$\mathcal{O}\left(nd\log d + \frac{n}{\epsilon^2}\log^3\frac{n}{\delta} + \frac{n^2}{\epsilon^2}\log\frac{n}{\delta}\right).$$

Answer:

By Lemma 5.50, let $\boldsymbol{A} \in \mathbb{R}^{k\times d}$ be a Gaussian random matrix, where every entry of $\boldsymbol{A}$ is sampled independently from $\mathcal{N}(0, 1/k)$. For every fixed vector $\boldsymbol{x} \in \mathbb{R}^d \setminus \{\boldsymbol{0}\}$ and $\epsilon_0 \in (0,1)$, we have

$$\Pr\left[\left|\frac{\|\boldsymbol{A}\boldsymbol{x}\|_2^2}{\|\boldsymbol{x}\|_2^2} - 1\right| \geq \epsilon_0\right] \leq 2e^{-k\epsilon_0^2/8}.$$

Following the proof of Theorem 5.49 (the JL lemma), for every $\epsilon_0 \in (0,1)$ and any integer k with

$$k \geq \frac{8(2\log n - \log\delta)}{\epsilon_0^2},$$

and the Gaussian random matrix $\boldsymbol{A}$ defined above, with probability at least $1-\delta$, for all $1 \leq i, j \leq n$,

$$(1-\epsilon_0)\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2 \leq \|\boldsymbol{A}(\boldsymbol{x}_i - \boldsymbol{x}_j)\|_2^2 \leq (1+\epsilon_0)\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2.$$

The algorithm proceeds as follows: first sample the Gaussian random matrix $\boldsymbol{A}$, then project all points to the k-dimensional space by computing $\boldsymbol{y}_i = \boldsymbol{A}\boldsymbol{x}_i$ for each i. Next apply the naïve algorithm to the compressed points $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n$ to find the pair with maximum distance, and output the corresponding original point pair. The total time complexity is $\mathcal{O}(nkd)$ for projecting all points plus $\mathcal{O}(n^2k)$ for computing all pairwise distances in the compressed space. Substituting the lower bound for k, this becomes

$$\mathcal{O}\left(\frac{n(n+d)}{\epsilon_0^2}\log\frac{n}{\delta}\right).$$

We now analyze the value of ϵ_0. To achieve the desired $1-\epsilon$ approximation guarantee, we require that

$$(1-\epsilon)^2 \leq \frac{1-\epsilon_0}{1+\epsilon_0} \implies \epsilon_0 \leq \frac{2\epsilon - \epsilon^2}{2 - 2\epsilon + \epsilon^2}.$$

Setting $\epsilon_0 = \frac{2\epsilon-\epsilon^2}{2-2\epsilon+\epsilon^2}$, the time complexity becomes

$$\mathcal{O}\left(\frac{n(n+d)}{\epsilon^2}\log\frac{n}{\delta}\right),$$

which matches the required bound.

Problem 5.16

Show that the absolute value in the definition of total variation distance is unnecessary. That is, for any two probability distributions P and Q,

$$d_{\mathrm{TV}}(P,Q) = \sup_{A \subseteq \mathbb{R}^d} \Big(\Pr[X \in A] - \Pr[Y \in A] \Big).$$

Note: Similarly, the absolute value in the variational form of total variation distance is also unnecessary:

$$d_{\mathrm{TV}}(P,Q) = \frac{1}{u-\ell} \sup_{\phi \in \mathcal{F}_{\ell,u}} \Big(P(\phi) - Q(\phi) \Big).$$

Answer:

Since $\Pr[X \in \mathbb{R}^d] = \Pr[Y \in \mathbb{R}^d] = 1$, we obtain

$$\Pr[X \in A] - \Pr[Y \in A] = -\big(\Pr[X \in (\mathbb{R}^d \setminus A)] - \Pr[Y \in (\mathbb{R}^d \setminus A)] \big).$$

The supremum of positive differences equals the supremum of absolute differences, so the absolute value is unnecessary.

Problem 5.17

Let P be a probability distribution on $\mathbb{R}$ and $X_1, \ldots, X_n$ be n i.i.d. samples from P. Define the empirical distribution to be the following discrete probability distribution on $\mathbb{R}$:

$$\hat{P}^{(n)}(A) := \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}_{[X_i \in A]}.$$

(a) Let P be a discrete probability distribution with PMF $p(x)$. Define $C(p) := \frac{1}{2} \sum_x \sqrt{p(x)(1-p(x))}$ and assume that $C(p) < +\infty$. Prove the following upper bound on $\mathbb{E}[d_{\mathrm{TV}}(\hat{P}^{(n)}, P)]$:

$$\mathbb{E}\left[d_{\mathrm{TV}}(\hat{P}^{(n)}, P) \right] \leq \frac{C(p)}{\sqrt{n}}.$$

(b) In the setting of the previous part, prove that for any $\epsilon > 0$ and $n \geq 1$,

$$\Pr\left[d_{\mathrm{TV}}(\hat{P}^{(n)}, P) \leq \frac{C(p)}{\sqrt{n}} + \epsilon \right] \geq 1 - e^{-2\epsilon^2 n}.$$

(c) If P is a continuous probability distribution, prove that for all $n \geq 1$, with probability 1,

$$d_{\mathrm{TV}}(\hat{P}^{(n)}, P) = 1.$$

Note: If we use the Wasserstein distance instead of the total variation distance, then for any probability distribution P on $[0,1]$,

$$\Pr\left[W_1(\hat{P}^{(n)}, P) \leq \epsilon \right] \geq 1 - 2e^{-2\epsilon^2 n}.$$

This follows from the Dvoretzky–Kiefer–Wolfowitz inequality.

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Answer:

For part (a), let $\hat{p}(x)$ be the PMF of $\hat{P}^{(n)}$. Let $C_x = \sum_{i=1}^{n} \mathbb{1}_{[X_i=x]}$ denote the random variable counting the number of occurrences of x, then $C_x \sim \text{Binomial}(n, p(x))$. By the linearity of expectation, we have

$$\begin{aligned}
\mathbb{E}\left[d_{\text{TV}}(\hat{P}^{(n)}, P)\right] &= \frac{1}{2}\sum_{x\in\mathcal{X}} \mathbb{E}\left[|\hat{p}(x) - p(x)|\right] \\
&= \frac{1}{2n}\sum_{x\in\mathcal{X}} \mathbb{E}\left[|C_x - np(x)|\right] \\
&\le \frac{1}{2n}\sum_{x\in\mathcal{X}} \sqrt{\mathbb{E}\left[\left(C_x - np(x)\right)^2\right]} \\
&= \frac{1}{2n}\sum_{x\in\mathcal{X}} \sqrt{\text{Var}(C_x)} \\
&= \frac{1}{2n}\sum_{x\in\mathcal{X}} \sqrt{np(x)(1-p(x))} \le \frac{C(p)}{\sqrt{n}}.
\end{aligned}$$

Here, we applied Jensen's inequality to $\mathbb{E}\left[|C_x - np(x)|\right]$ using the concave function $f(x) = \sqrt{x}$, and used the variance formula for the binomial random variable C_x.

For part (b), let $f(X_1, \ldots, X_n) = d_{\text{TV}}(\hat{P}^{(n)}, P)$, where $\hat{P}^{(n)}$ is the empirical distribution of $X_1, \ldots, X_n$. We now show that for each $i = 1, \ldots, n$, any fixed $x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n$, and any $x_i, x_i' \in \mathcal{X}$,

$$|f(x_1, \ldots, x_i, \ldots, x_n) - f(x_1, \ldots, x_i', \ldots, x_n)| \le \frac{1}{n}.$$

This follows from the triangle inequality $|d_{\text{TV}}(\hat{P}^{(n)}, P) - d_{\text{TV}}(\hat{P}'^{(n)}, P)| \le d_{\text{TV}}(\hat{P}^{(n)}, \hat{P}'^{(n)})$. The two empirical distributions differ only at the points x_i and x_i', each by a probability mass of $1/n$, which implies $d_{\text{TV}}(\hat{P}^{(n)}, \hat{P}'^{(n)}) = \frac{1}{2} \cdot \frac{2}{n} = \frac{1}{n}$.

Therefore, by McDiarmid's inequality, for all $\epsilon \ge 0$,

$$\Pr\left[f(X_1, \ldots, X_n) \ge \mathbb{E}[f(X_1, \ldots, X_n)] + \epsilon\right] \le e^{-2\epsilon^2/L^2},$$

where $L^2 = \sum_{i=1}^{n} (1/n)^2 = 1/n$. Substituting $\mathbb{E}[f(X_1, \ldots, X_n)] \le C(p)/\sqrt{n}$ and taking the complement event, we obtain

$$\Pr\left[d_{\text{TV}}(\hat{P}^{(n)}, P) \le \frac{C(p)}{\sqrt{n}} + \epsilon\right] \ge 1 - e^{-2\epsilon^2 n}.$$

For part (c), since $\hat{P}^{(n)}$ is a discrete distribution, the total variation distance between $\hat{P}^{(n)}$ and the continuous distribution P is

$$d_{\text{TV}}(\hat{P}^{(n)}, P) = \sup_{A\subseteq\mathbb{R}} \left|\hat{P}^{(n)}(A) - P(A)\right|.$$

Let $S = \{X_i \mid 1 \le i \le n\}$ be the set of sample points. Taking the set $A = \mathbb{R} \setminus S$, we have $\hat{P}^{(n)}(A) = 0$ and $P(A) = 1$ almost surely, since only finitely many points are removed from $\mathbb{R}$. Therefore,

$$d_{\text{TV}}(\hat{P}^{(n)}, P) = 1.$$

Problem 5.18

The Gaussian distribution also has an important application in Fourier analysis. In this exercise, we will learn a little bit about the Fourier transform and its inverse. The Fourier transform of a function $f : \mathbb{R} \to \mathbb{C}$ is defined as the function $\hat{f} : \mathbb{R} \to \mathbb{C}$ given by

$$\hat{f}(\omega) := \int_{-\infty}^{+\infty} e^{i\omega x} f(x)\,\mathrm{d}x.$$

The inverse Fourier transform of a function $\hat{f} : \mathbb{R} \to \mathbb{C}$ is defined as the function $\check{f} : \mathbb{R} \to \mathbb{C}$ given by

$$\check{f}(x) := \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-i\omega x} \hat{f}(\omega)\,\mathrm{d}\omega.$$

We denote the Fourier transform of f by $\mathcal{F}[f]$ and the inverse Fourier transform of $\hat{f}$ by $\mathcal{F}^{-1}[\hat{f}]$. A sufficient condition for the Fourier transform of f to be well-defined is that $f \in L^1(\mathbb{R})$, where $L^1(\mathbb{R})$ is the space of functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$\int_{-\infty}^{+\infty} |f(x)|\,dx < +\infty.$$

(a) Let φ_σ be the PDF of the Gaussian distribution $\mathcal{N}(0, \sigma^2)$. Compute its Fourier transform $\mathcal{F}[\varphi_\sigma]$ and verify that

$$\mathcal{F}^{-1}\left[\mathcal{F}[\varphi_\sigma]\right] = \varphi_\sigma.$$

(b) Let $f \in L^1(\mathbb{R})$. For every $\sigma > 0$, define the Gaussian smoothing of f with bandwidth σ to be the function $f_\sigma : \mathbb{R} \to \mathbb{R}$ given by

$$f_\sigma(x) := \mathbb{E}[f(x + Z)] = \int_{-\infty}^{+\infty} f(x + z)\varphi_\sigma(z)\,\mathrm{d}z,$$

where $Z \sim \mathcal{N}(0, \sigma^2)$. Prove that $f_\sigma \in L^1(\mathbb{R})$.

(c) Express $\mathcal{F}[f_\sigma]$ in terms of $\mathcal{F}[f]$ and verify that

$$\mathcal{F}^{-1}\left[\mathcal{F}[f_\sigma]\right] = f_\sigma.$$

(d) Let $f \in L^1(\mathbb{R})$ be a function such that both the left and right limits of f exist at every point in $\mathbb{R}$:

$$f_+(x) := \lim_{y \to x^+} f(y), \quad f_-(x) := \lim_{y \to x^-} f(y).$$

Prove that

$$\mathcal{F}^{-1}[\mathcal{F}[f]] = \frac{f_- + f_+}{2}.$$

Although this problem looks more rigorous about integration than this course usually requires, we still allow you to swap the order of the limit and integration without justification.

(e) Let X, Y be two random variables with continuous PDFs f_X and f_Y. Prove that if $M_X(it) = M_Y(it)$ for all $t \in \mathbb{R}$, then $f_X = f_Y$.

Note: This provides an alternative proof for Remark 4.91 in the continuous case.

Answer:

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For part (a), we first write down the expression for φ_σ:

$$\varphi_\sigma(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/(2\sigma^2)}.$$

Comparing the definition of the Fourier transform with that of the moment generating function, we obtain

$$\mathcal{F}[\varphi_\sigma](\omega) = \int_{-\infty}^{+\infty} e^{i\omega x}\varphi_\sigma(x)\,\mathrm{d}x = M_X(i\omega) = e^{\sigma^2(i\omega)^2/2} = e^{-\sigma^2\omega^2/2},$$

where $M_X(t) = e^{\sigma^2 t^2/2}$ is the moment generating function of a Gaussian random variable $X \sim \mathcal{N}(0, \sigma^2)$. For the inverse Fourier transform, we compare the integral form with the result we just derived:

$$\int_{-\infty}^{+\infty} e^{it\omega} e^{-\sigma^2\omega^2/2}\,\mathrm{d}\omega = \sqrt{\frac{2\pi}{\sigma^2}} e^{-t^2/(2\sigma^2)}.$$

Setting $t = -x$ in the above identity, we get

$$\mathcal{F}^{-1}\left[\mathcal{F}[\varphi_\sigma]\right](x) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} e^{-i\omega x}\mathcal{F}[\varphi_\sigma](\omega)\,\mathrm{d}\omega = \frac{1}{2\pi}\cdot\sqrt{\frac{2\pi}{\sigma^2}} e^{-(-x)^2/(2\sigma^2)} = \frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/(2\sigma^2)}.$$

Therefore, we have verified that

$$\mathcal{F}^{-1}\left[\mathcal{F}[\varphi_\sigma]\right] = \varphi_\sigma.$$

For part (b), let $I = \int_{-\infty}^{+\infty} |f(x)|\,\mathrm{d}x$, then

$$\begin{aligned}
\int_{-\infty}^{+\infty} |f_\sigma(x)|\,\mathrm{d}x &= \int_{-\infty}^{+\infty} |\mathbb{E}[f(x+Z)]|\,\mathrm{d}x \\
&\le \mathbb{E}\left[\int_{-\infty}^{+\infty} |f(x+Z)|\,\mathrm{d}x\right] \\
&= \mathbb{E}[I] = I.
\end{aligned}$$

Hence, $f_\sigma \in L^1(\mathbb{R})$.

For part (c), by definition, we have

$$\begin{aligned}
\mathcal{F}[f_\sigma](\omega) &= \int_{-\infty}^{+\infty} e^{i\omega x} f_\sigma(x)\,\mathrm{d}x \\
&= \int_{-\infty}^{+\infty} e^{i\omega x}\int_{-\infty}^{+\infty} f(x+z)\varphi_\sigma(z)\,\mathrm{d}z\,\mathrm{d}x \\
&= \int_{-\infty}^{+\infty} \varphi_\sigma(z)\int_{-\infty}^{+\infty} e^{i\omega x} f(x+z)\,\mathrm{d}x\,\mathrm{d}z \\
&= \int_{-\infty}^{+\infty} e^{-i\omega z}\varphi_\sigma(z)\int_{-\infty}^{+\infty} e^{i\omega(x+z)} f(x+z)\,\mathrm{d}x\,\mathrm{d}z \\
&= \int_{-\infty}^{+\infty} e^{-i\omega z}\varphi_\sigma(z)\mathcal{F}[f](\omega)\,\mathrm{d}z \\
&= \mathcal{F}[f](\omega)\cdot\int_{-\infty}^{+\infty} e^{i(-\omega)z}\varphi_\sigma(z)\,\mathrm{d}z \\
&= \mathcal{F}[f](\omega)\cdot\mathcal{F}[\varphi_\sigma](-\omega) = e^{-\sigma^2\omega^2/2}\mathcal{F}[f](\omega).
\end{aligned}$$

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For the inverse transform, we have

$$\begin{aligned}
\mathcal{F}^{-1}\left[\mathcal{F}[f_\sigma]\right](x) &= \frac{1}{2\pi}\int_{-\infty}^{+\infty} e^{-i\omega x}\mathcal{F}[f_\sigma](\omega)\,\mathrm{d}\omega \\
&= \frac{1}{2\pi}\int_{-\infty}^{+\infty} e^{-i\omega x}e^{-\sigma^2\omega^2/2}\int_{-\infty}^{+\infty} e^{i\omega t}f(t)\,\mathrm{d}t\,\mathrm{d}\omega \\
&= \int_{-\infty}^{+\infty} f(t)\int_{-\infty}^{+\infty}\frac{1}{2\pi}e^{-i\omega(x-t)}e^{-\sigma^2\omega^2/2}\,\mathrm{d}\omega\,\mathrm{d}t \\
&= \int_{-\infty}^{+\infty} f(t)\varphi_\sigma(t-x)\,\mathrm{d}t \\
&= \int_{-\infty}^{+\infty} f(x+z)\varphi_\sigma(z)\,\mathrm{d}z = f_\sigma(x).
\end{aligned}$$

Therefore, we have verified that

$$\mathcal{F}^{-1}\left[\mathcal{F}[f_\sigma]\right] = f_\sigma.$$

For part (d), as $\sigma \to 0$, we have

$$\mathcal{F}[f_\sigma](\omega) = e^{-\sigma^2\omega^2/2}\mathcal{F}[f](\omega) \to \mathcal{F}[f](\omega),$$

which implies that

$$f_\sigma = \mathcal{F}^{-1}\left[\mathcal{F}[f_\sigma]\right] \to \mathcal{F}^{-1}\left[\mathcal{F}[f]\right].$$

By definition, we have

$$\begin{aligned}
f_\sigma(x) &= \int_{-\infty}^{+\infty} f(x+z)\varphi_\sigma(z)\,\mathrm{d}z \\
&= \int_{-\infty}^{+\infty} f(x+z)\frac{1}{\sigma\sqrt{2\pi}}e^{-z^2/(2\sigma^2)}\,\mathrm{d}z \\
&= \int_{-\infty}^{+\infty} f(x+\sigma z)\frac{1}{\sqrt{2\pi}}e^{-z^2/2}\,\mathrm{d}z \\
&= \int_{-\infty}^{0} f(x+\sigma z)\varphi_1(z)\,\mathrm{d}z + \int_{0}^{+\infty} f(x+\sigma z)\varphi_1(z)\,\mathrm{d}z.
\end{aligned}$$

For any fixed $z \geq 0$, as $\sigma \to 0$, we have $f(x+\sigma z) \to f_+(x)$, and similarly for $z \leq 0$, $f(x+\sigma z) \to f_-(x)$. Since $\int_{-\infty}^{0}\varphi_1(z)\,\mathrm{d}z = \int_{0}^{+\infty}\varphi_1(z)\,\mathrm{d}z = \frac{1}{2}$, we obtain

$$\lim_{\sigma\to 0} f_\sigma(x) = \frac{f_-(x)+f_+(x)}{2} \implies \mathcal{F}^{-1}\left[\mathcal{F}[f]\right] = \frac{f_-+f_+}{2}.$$

For part (e), notice that $M_X(it) = \int_{-\infty}^{+\infty} e^{itx}f_X(x)\,\mathrm{d}x = \hat{f}_X(t)$, so $M_X(it) = M_Y(it)$ implies $\mathcal{F}[f_X] = \mathcal{F}[f_Y]$. For a continuous function f, we have $f_-(x) = f_+(x) = f(x)$ pointwise. Therefore, applying the inverse Fourier transform to both sides of $\mathcal{F}[f_X] = \mathcal{F}[f_Y]$, we obtain $f_X = f_Y$.